Proof. For $n = 0$, there is nothing to prove. Assume the goal holds for some $n \geq 0$. For $x, y \in R$, set $u = x^{2^n}$, so that $\mathrm{ad}(u) = \mathrm{ad}(x)^{2^n}$. Then

$$\begin{aligned}
\mathrm{ad}(u^2)(y) &= [u^2, y] = u^2 y - yu^2 \\
&= u^2 y - uyu + uyu - yu^2 \\
&= [u, uy] + [u, yu] \\
&= \mathrm{ad}(u)(uy) + \mathrm{ad}(u)(yu) \\
&= \mathrm{ad}(u)(uy) - \mathrm{ad}(u)(yu) \\
&= \mathrm{ad}(u)([u, y]) \\
&= \mathrm{ad}(u)^2(y)\,.
\end{aligned}$$

Thus

$$\mathrm{ad}(x^{2^{n+1}})(y) = \mathrm{ad}(u^2) = \mathrm{ad}(u)^2 = (\mathrm{ad}(x)^{2^n})^2(y) = \mathrm{ad}(x)^{2^{n+1}}(y)\,.$$

By induction, we have the desired result. □

Lemma 3.3. *Let R be a ring satisfying (E). Then for all $x, y \in R$,*

$$[x, [x, y]] = 0 \implies [x, y] = 0\,,$$

or equivalently,

$$\mathrm{ad}(x)^2(y) = 0 \implies \mathrm{ad}(x)(y) = 0\,.$$

Proof. Assume $[x, [x, y]] = \mathrm{ad}(x)^2(y) = 0$ for some $x, y \in R$. By Lemmas 3.1 and 3.2, $\mathrm{ad}(x) = \mathrm{ad}(p(x)) = p(\mathrm{ad}(x))$. Since $p(t) = q(t^2)t^2$ for some $q(t) \in \mathbb{Z}[t]$, it follows that $\mathrm{ad}(x)(y) = q(\mathrm{ad}(x)^2)\mathrm{ad}(x)^2(y) = 0$. □

Lemma 3.4. *Let R be a ring satisfying (E). Then for all $x, y \in R$,*

$$[x, [x, y]] = [x, y] \implies [x, y] = 0\,,$$

or equivalently,

$$\mathrm{ad}(x)^2(y) = \mathrm{ad}(x)(y) \implies \mathrm{ad}(x)(y) = 0\,.$$

Proof. Let $x, y \in R$ satisfy $[x, [x, y]] = [x, y]$. Then

$$[[x, y], [[x, y], x]] = -[[x, y], [x, [x, y]]] = -[[x, y], [x, y]] = 0\,.$$

By Lemma 3.3, $[[x, y], x] = 0$. Thus $[x, y] = [x, [x, y]] = 0$. □

Theorem 3.5. *Let R be a ring satisfying $[x^2 - x, y] = 0$ for all $x, y \in R$. Then R is commutative.*

Proof. Since (E) holds, Lemmas 3.1 and 3.2 give $\mathrm{ad}(x)(y) = \mathrm{ad}(x^2)(y) = \mathrm{ad}(x)^2(y)$ for all $x, y \in R$. By Lemma 3.4, $\mathrm{ad}(x)(y) = 0$ for all $x, y \in R$, that is, R is commutative. (See also Lemma 2.9.) □

Lemma 3.6. *Let R be a ring, let n be a positive integer and assume $[x^{2^n} - x, y] = 0$ for all $x, y \in R$. Then $[x^{2^{n-1}} + \cdots + x^2 + x, y] = 0$ for all $x, y \in R$.*

Proof. For all $x \in R$, $\mathrm{ad}(x)^{2^n} = \mathrm{ad}(x^{2^n}) = \mathrm{ad}(x)$, using Lemmas 3.1 and 3.2. Now for all $x \in R$ (and using Lemma 3.1),

$$\begin{aligned}
\mathrm{ad}(x^{2^{n-1}} + \cdots + x^2 + x)^2 &= (\mathrm{ad}(x^{2^{n-1}}) + \cdots + \mathrm{ad}(x^2) + \mathrm{ad}(x))^2 \\
&= (\mathrm{ad}(x)^{2^{n-1}} + \cdots + \mathrm{ad}(x)^2 + \mathrm{ad}(x))^2 \\
&= \mathrm{ad}(x)^{2^n} + \mathrm{ad}(x)^{2^{n-1}} + \cdots + \mathrm{ad}(x)^4 + \mathrm{ad}(x)^2 \\
&= \mathrm{ad}(x) + \mathrm{ad}(x)^{2^{n-1}} + \cdots + \mathrm{ad}(x)^4 + \mathrm{ad}(x)^2 \\
&= \mathrm{ad}(x^{2^{n-1}}) + \cdots + \mathrm{ad}(x^2) + \mathrm{ad}(x) \\
&= \mathrm{ad}(x^{2^{n-1}} + \cdots + x^2 + x)
\end{aligned}$$

By Lemma 3.4, $\mathrm{ad}(x^{2^{n-1}} + \cdots + x^2 + x) = 0$. This proves the desired result. □

Theorem 3.7. *Let R be a ring satisfying $[x^4 - x, y] = 0$ for all $x, y \in R$. Then R is commutative.*

Proof. This follows from taking $n = 2$ in Lemma 3.6 and using Theorem 3.5. □

We conclude with the result that began this whole endeavor. We are well aware that of all the proofs in this paper, this is the one that most seems like it was generated by an automated deduction tool. We have simplified the proof to the point where it is possible to follow each individual step, but we would certainly agree that it is difficult to see how a human would have found the proof.

Theorem 3.8. *Let R be a ring satisfying $[x^8 - x, y] = 0$ for all $x, y \in R$. Then R is commutative.*

Proof. We will freely use Lemma 3.1 without explicit reference. By Lemma 3.6, $[x^4 + x^2 + x, y] = 0$ for all $x, y \in R$, that is, $\mathrm{ad}(x)^4 + \mathrm{ad}(x)^2 + \mathrm{ad}(x) = 0$.

First, for all $u, v \in R$,

$$\begin{aligned}
(\mathrm{ad}(u)^2 + \mathrm{ad}(v)^2)([u,v]) &= (\mathrm{ad}(u)^2 + \mathrm{ad}([u,v]) + \mathrm{ad}(v)^2)([u,v]) \\
&= (\mathrm{ad}(u)^2 + \mathrm{ad}(u)\mathrm{ad}(v) + \mathrm{ad}(v)\mathrm{ad}(u) + \mathrm{ad}(v)^2)([u,v]) \\
&= (\mathrm{ad}(u) + \mathrm{ad}(v))^2([u,v]) \\
&= \mathrm{ad}(u+v)^2([u,v]) \\
&= \mathrm{ad}(u+v)^2([u+v,v]) \\
&= \mathrm{ad}(u+v)^3(v)\,.
\end{aligned}$$

Thus

$$\begin{aligned}
\mathrm{ad}(u+v)(\mathrm{ad}(u)^2 + \mathrm{ad}(v)^2)([u,v]) &= \mathrm{ad}(u+v)^4(v) \\
&= \mathrm{ad}(u+v)^2(v) + \mathrm{ad}(u+v)(v) \\
&= \mathrm{ad}(u+v)([u+v,v] + v) \\
&= \mathrm{ad}(u+v)(v + [u,v])\,.
\end{aligned}$$

Rearranging this, we have $\mathrm{ad}(u+v)\mathrm{ad}(v)^2([u,v]) = \mathrm{ad}(u+v)(v + [u,v] + \mathrm{ad}(u)^2([u,v]))$, that is,

$$\mathrm{ad}(u+v)\mathrm{ad}(v)^3(u) = \mathrm{ad}(u+v)(v + \mathrm{ad}(u)(v) + \mathrm{ad}(u)^3(v))\,.$$

In this last equation, replace v with $[u,v]$ to get

$$[u + [u,v], \mathrm{ad}([u,v])^3(u)] = [u + [u,v], \mathrm{ad}(u)(v) + \mathrm{ad}(u)^2(v) + \mathrm{ad}(u)^4 v] = 0\,. \tag{6}$$